

Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
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Laporan Akhir

Praktikum Jaringan Komputer

Vlan-Trunk-OSPF-MultiVendor

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2026

1 Langkah-Langkah Percobaan

1.1 Yang pertama

1. Masuk ke terminal Cisco Switch

Switch> enable

Switch configure terminal

Switch(config)

2. Membuat Vlan

vlan 10

name Finance

exit

vlan 20

name HR

exit

vlan 30

name IT

exit

vlan 40

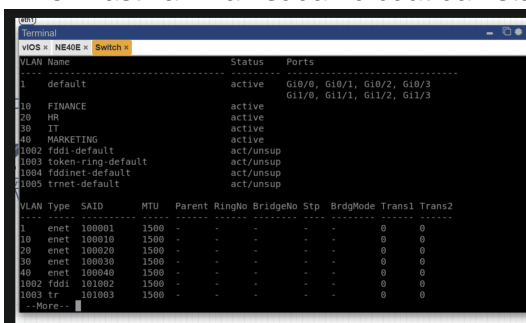
name Marketingg

exit

Verifikasi:

show vlan brief

3. Pastikan vlan sudah dibuat dan statusnya aktif



VLAN	Name	Status	Ports
1	default	active	Gi0/8, Gi0/1, Gi0/2, Gi0/3
10	FINANCE	active	Gi1/8, Gi1/1, Gi1/2, Gi1/3
20	HR	active	
30	IT	active	
40	MARKETING	active	
1002	fdi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fdi-default	act/unsup	
1005	trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BridgeMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
10	enet	100010	1500	-	-	-	-	-	0	0
20	enet	100020	1500	-	-	-	-	-	0	0
30	enet	100030	1500	-	-	-	-	-	0	0
40	enet	100040	1500	-	-	-	-	-	0	0
1002	fdi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0

4. Konfigurasi Access port untuk setiap vlan

Port Access harus disesuaikan dengan interface pada cisco switch

Switchconf t

interface gi0/2

```
description PC-FINANCE-VLAN10
switchport mode access
switchport access vlan 10
no shutdown
exit
```

```
interface gi0/1
description PC-HR-VLAN20
switchport mode access
switchport access vlan 20
no shutdown
exit
```

```
interface gi1/1
description PC-IT-VLAN30
switchport mode access
switchport access vlan 30
no shutdown
exit
```

```
interface gi0/3
description PC-MARKETING-VLAN40
switchport mode access
switchport access vlan 40
no shutdown
exit
```

5. Konfigurasi Trunk ke Cisco Router

Konfigurasi trunk port dari switch menuju cisco router untuk vlan 10, 20, 30, dan 40.
Switchconf t

```
interface gi0/0
switchport trunk encapsulation dot1q
description TRUNK-TO-CISCO-ROUTER
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40
no shutdown
exit
```

```
show interfaces trunk
```

```
write memory
```

```

Port      Mode      Encapsulation  Status  Native vlan
Gi0/0     on        802.1q         trunking  1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#

```

6. Hasil yang diharapkan

Praktikum 1 selesai jika konfigurasi memenuhi berikut:

Vlan 10, 20, 30, dan 40 sudah ada.

Trunk port sudah membawa vlan 10, 20, 30, dan 40.

Access port setiap vlan sudah sesuai.

1.2 yang kedua

1. Masuk ke Cisco Router

enable

configure terminal

2. Konfigurasi Interface Trunk ke Switch

Pilih interface yang terhubung ke Cisco Switch. Pada topologi, interface yang terhubung adalah Gig0/2

```
interface gig0/2
```

```
description TRUNK-TO-CISCO-SWITCH
```

```
no ip address
```

```
no shutdown
```

```
exit
```

3. Konfigurasi Subinterface VLAN 10

```
interface gig0/2.10
```

```
description GATEWAY-VLAN10-FINANCE
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

4. Konfigurasi Subinterface VLAN 20

```
interface gig0/2.20
```

```
description GATEWAY-VLAN20-HR
```

```
encapsulation dot1Q 20
```

```
ip address 192.168.20.1 255.255.255.0
```

```
no shutdown
```

```
exit
```

5. Konfigurasi Subinterface VLAN 30

```
interface gig0/2.30
```

```
description GATEWAY-VLAN30-IT
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

6. Konfigurasi Subinterface VLAN 40

```
interface gig0/2.40
description GATEWAY-VLAN40-MARKETING
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
```

7. Konfigurasi DHCP-Server untuk Vlan 10

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
```

```
ip dhcp pool VLAN10-FINANCE
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
```

8. Konfigurasi DHCP-Server untuk Vlan 20

```
ip dhcp excluded-address 192.168.20.1 192.168.20.10
```

```
ip dhcp pool VLAN20-HR
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit
```

9. Konfigurasi IP Static pada VPCS untuk vlan 30 dan 40

PC Vlan 30 :

```
ip 192.168.30.10/24 192.168.30.1
```

PC Vlan 40 :

```
ip 192.168.40.10/24 192.168.40.1
```

PC Vlan 10 dan 20 menggunakan dhcp:

```
ip dhcp
```

10. Verifikasi Cisco Router

Cek interface:

```
show ip dhcp binding
```

The image displays two terminal windows side-by-side, showing the configuration and status of a Cisco switch.

Left Terminal (Configuration):

```

VSW1# NS44# Switch > IT > MARKETING > HR > FINANCE >
Router(config)#dns server 8.8.8.8
Router(config)#exit
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(config)#network 192.168.10.1
Router(config)#exit
Router(config)#interface gig0/20
Router(config)#description GATEWAY-VLAN10-FINANCE
Router(config)#ip dhcp relay address 192.168.10.1 255.255.255.0
Router(config)#ip dhcp relay shutdown
Router(config)#exit
Router(config)#exit
Router#show ip
Jun  6 05:40:09.798: SYS-5-CMG1-1: Configured from console by consolein
Adminlog command: "show ip"
Router#show ip interface brief

```

Right Terminal (Status):

```

VSW1# NS44# Switch > IT > MARKETING > HR > FINANCE >
 gigabitEthernet0/20 192.168.20.1 YES manual up up
 gigabitEthernet0/20 192.168.30.1 YES manual up up
 gigabitEthernet0/20 192.168.40.1 YES manual up up
 gigabitEthernet0/30 unassigned YES unset administratively down down
Router#show ip dhcp pool
Pool VLAN10-FINANCE:
  Utilization MARK (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
  Current index IP address range Leased addresses
  192.168.10.12 192.168.10.1 - 192.168.20.254 1
Pool VLAN20-HR:
  Utilization MARK (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
  Current index IP address range Leased addresses
  192.168.20.12 192.168.20.1 - 192.168.20.254 1
Router#

```

11. Pengujian dari VPCS

Masuk ke pc vlan 10 untuk request ip secara dhcp:

```
ip dhcp
```

Pastikan pc vlan 10 sudah dapat ip secara dhcp dengan command:

show ip

Dari pc vlan 10 ping gateway 192.168.10.1:

```
ping 192.168.10.1
```

Masuk ke pc vlan 20 untuk request ip secara dhcp:

```
ip dhcp
```

Pastikan pc vlan 20 sudah dapat ip secara dhcp dengan command:

show ip

Dari pc vlan 20 ping gateway 192.168.20.1:

```
ping 192.168.20.1
```

Dari pc vlan 30 ping gateway 192.168.30.1:

```
ping 192.168.30.1
```

```
ping 192.168.40.1
```

Uji ping antar-Vlan dari vlan 10:

```
ping 192.168.20.11
```

```
ping 192.168.30.10
```

```
ping 192.168.40.10
```

Masuk ke cisco router lagi untuk melihat ip dhcp binding:

```
show ip dhcp binding
```

[illegible]

```
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.578 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.034 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.688 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.273 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=4.112 ms

VPCS> █
```

```

Terminal
VDS > NE40E > Switch > IT > MARKETNO > HR > FINANCE >

VPCS> ip dhcp
SDRA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show ip

NAME      : VPCS[1]
IP/MASK   : 192.168.20.11/24
GATEWAY   : 192.168.20.1
DNS       : 8.8.8.8
DHCP SERVER: 192.168.20.1
DHCP LEASE: 86129, 86400/43200/75600
MAC       : 08:100:79:66:60:2d
PORT      : 20000
HOST:PORT : 127.0.0.1:30000
RTU        : 1500

VPCS> ping 192.168.20.1

44 bytes from 192.168.20.1: icmp_seq=1 ttl=255 time=3.507 ms
44 bytes from 192.168.20.1: icmp_seq=2 ttl=255 time=3.461 ms
44 bytes from 192.168.20.1: icmp_seq=3 ttl=255 time=3.822 ms
44 bytes from 192.168.20.1: icmp_seq=4 ttl=255 time=4.233 ms
44 bytes from 192.168.20.1: icmp_seq=5 ttl=255 time=4.792 ms

VPCS>

```

```
VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=2.260 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.199 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=5.010 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=5.346 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=3.325 ms

VPCS> █
```

[illegible]

```

# Bindings from all pools not associated with VRF:
IP address      Hardware address/  Lease expiration  Type
                Client ID/
192.168.10.11    0100.5079.6668.2c  Jun 07 2026 05:48 AM  Automatic
192.168.20.11    0100.5079.6668.2d  Jun 07 2026 05:46 AM  Automatic
Router#

```

1.3 yang ketiga

1. Konfigurasi Lan Huawei

konfigurasi ip address untuk lan huawei:

system-view

```

interface Ethernet1/0/2
description LAN-HUAWEI
ip address 192.168.50.1 255.255.255.0
undo shutdown
quit

```

```
commit
```

2. Konfigurasi IP VPC Lan Huawei

Pada pc lan huawei:

```
ip 192.168.50.10/24 192.168.50.1
```

Verifikasi:

```
show ip
```

Testing ping gateway huawei:

```
ping 192.168.50.1
```

```

CHECKING FOR duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:4b
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS>

```

```

VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=15.015 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=2.436 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.871 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.959 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.008 ms

VPCS>

```

3. Konfigurasi IP link cisco <-> Huawei link 1

Network:

```
10.10.10.0/30
```

Cisco Router

```
enable
```

configure terminal

```

interface gi0/3
description LINK1-TO-HUAWEI
ip address 10.10.10.1 255.255.255.252
no shutdown
exit
Huawei Router

```



```
system-view
```

```
interface Ethernet1/0/0  
description LINK1-TO-CISCO  
ip address 10.10.10.2 255.255.255.252  
undo shutdown  
quit
```

```
commit
```

4. Konfigurasi IP Link Cisco Huawei Link 2
Network:

10.10.11.0/30

Cisco Router

```
interface gi0/0  
description LINK2-TO-HUAWEI  
ip address 10.10.11.1 255.255.255.252  
no shutdown  
exit  
Huawei Router
```

```
system-view
```

```
interface Ethernet1/0/4  
description LINK2-TO-CISCO  
ip address 10.10.11.2 255.255.255.252  
undo shutdown  
quit
```

```
commit
```

5. Konfigurasi IP link Huawei <-> Mikrotik
Network:

10.10.20.0/30

Huawei Router

```
system-view
```

```
interface Ethernet1/0/3
```

```
description LINK-TO-MIKROTIK
ip address 10.10.20.1 255.255.255.252
undo shutdown
quit
```

```
commit
```

Mikrotik

```
/ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
```

6. Konfigurasi IP link Cisco <-> Mikrotik Network:

10.10.30.0/30

Cisco Router

```
interface gi0/1
description LINK-TO-MIKROTIK
ip address 10.10.30.1 255.255.255.252
no shutdown
exit
Mikrotik
```

```
/ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
```

7. Verifikasi Interface

Cisco Router

```
show ip interface brief
```

Huawei Router

```
display ip interface brief
```

Mikrotik

```
/ip address print
```

```

% Ambiguous command: "show ip 1"
Router#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0  10.10.11.1      YES manual up       up
GigabitEthernet0/1  10.10.30.1      YES manual up       up
GigabitEthernet0/2  unassigned      YES unset  up       up
GigabitEthernet0/2.30  192.168.10.1   YES manual up       up
GigabitEthernet0/2.20  192.168.20.1   YES manual up       up
GigabitEthernet0/2.30  192.168.30.1   YES manual up       up
GigabitEthernet0/2.40  192.168.40.1   YES manual up       up
GigabitEthernet0/3    10.10.10.1      YES manual up       up
Router#

--HUAWEI--display ip interface brief
--down: administratively down
--down: FIB overload down
--down: standby
(U): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface      IP Address/Mask  Physical  Protocol  VPN
Ethernet1/0/0   10.10.10.2/30    up        up        --
Ethernet1/0/0.4094  128.1.138.137/16 up        up        13vpn
Ethernet1/0/1     unassigned       up        down      --
Ethernet1/0/1.4094  128.1.138.137/16 up        up        13vpn
Ethernet1/0/2     192.168.50.1/24 up        up        --
Ethernet1/0/2.4094  128.1.138.137/16 up        up        13vpn
Ethernet1/0/3     10.10.20.1/30    up        up        --
Ethernet1/0/3.4094  128.1.138.137/16 up        up        13vpn
Ethernet1/0/4     10.10.11.2/30    up        up        --
Ethernet1/0/4.4094  128.1.138.137/16 up        up        13vpn
---- More ----

[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;;; TO-HUAWEI
< 10.10.20.2/30 10.10.20.0 ether1
1 ;;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
[admin@MikroTik] >

```

8. Verifikasi ping Antar Router

Dari Cisco Router

ping 10.10.10.2

ping 10.10.11.2

ping 10.10.30.2

Target:

Cisco harus bisa ping ke huawei dan mikrotik.

Dari Router Huawei

ping 10.10.10.1

ping 10.10.11.1

ping 10.10.20.2

Target: Huawei harus bisa ping cisco dan mikrotik.

Dari Router Mikrotik

ping 10.10.20.1

ping 10.10.30.1

Target: Mikrotik harus bisa ping ke cisco dan huawei.

```

Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/2/5 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#

```

```

Terminal
VIOS x NE40E x Switch x MARKETING x IT x HR x FINANCE x VPC x Mikrotik x

[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.10.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=3 ms
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 3/3/3 ms

[~HUAWEI]

```

```

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=2 ms
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=3 ms
Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=2 ms

--- 10.10.20.2 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/2/3 ms

[~HUAWEI]

```

```

Terminal
VIOS x NE40E x Switch x MARKETING x IT x HR x FINANCE x VPC x Mikrotik x

admin@mikrotik:~$ ping 10.10.20.1
PING 10.10.20.1: 56 data bytes, press CTRL_C to break
0 10.10.20.1: 56 bytes=56 Sequence=1 ttl=64 time=2 ms
1 10.10.20.1: 56 bytes=56 Sequence=2 ttl=64 time=1 ms
2 10.10.20.1: 56 bytes=56 Sequence=3 ttl=64 time=2 ms
3 10.10.20.1: 56 bytes=56 Sequence=4 ttl=64 time=3 ms
4 10.10.20.1: 56 bytes=56 Sequence=5 ttl=64 time=2 ms
5 10.10.20.1: 56 bytes=56 Sequence=6 ttl=64 time=2 ms
6 10.10.20.1: 56 bytes=56 Sequence=7 ttl=64 time=2 ms
7 10.10.20.1: 56 bytes=56 Sequence=8 ttl=64 time=2 ms
8 10.10.20.1: 56 bytes=56 Sequence=9 ttl=64 time=2 ms
9 10.10.20.1: 56 bytes=56 Sequence=10 ttl=64 time=2 ms

--- 10.10.20.1 ping statistics ---
10 packet(s) transmitted
10 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/2/3 ms

admin@mikrotik:~$ ping 10.10.30.1
PING 10.10.30.1: 56 data bytes, press CTRL_C to break
0 10.10.30.1: 56 bytes=56 Sequence=1 ttl=64 time=2 ms
1 10.10.30.1: 56 bytes=56 Sequence=2 ttl=64 time=1 ms
2 10.10.30.1: 56 bytes=56 Sequence=3 ttl=64 time=2 ms
3 10.10.30.1: 56 bytes=56 Sequence=4 ttl=64 time=3 ms
4 10.10.30.1: 56 bytes=56 Sequence=5 ttl=64 time=2 ms
5 10.10.30.1: 56 bytes=56 Sequence=6 ttl=64 time=2 ms
6 10.10.30.1: 56 bytes=56 Sequence=7 ttl=64 time=2 ms
7 10.10.30.1: 56 bytes=56 Sequence=8 ttl=64 time=2 ms
8 10.10.30.1: 56 bytes=56 Sequence=9 ttl=64 time=2 ms
9 10.10.30.1: 56 bytes=56 Sequence=10 ttl=64 time=2 ms

--- 10.10.30.1 ping statistics ---
10 packet(s) transmitted
10 packet(s) received
0.00% packet loss
round-trip min/avg/max = 1/2/3 ms

admin@mikrotik:~$

```

1.4 yang keempat

3. Konfigurasi ospf pada Cisco Router

Masuk ke Cisco Router:

enable

configure terminal

Konfigurasi OSPF:

router ospf 1

router-id 1.1.1.1

network 10.10.10.0 0.0.0.3 area 0

network 10.10.11.0 0.0.0.3 area 0

network 10.10.30.0 0.0.0.3 area 0

network 192.168.10.0 0.0.0.255 area 0

network 192.168.20.0 0.0.0.255 area 0

network 192.168.30.0 0.0.0.255 area 0

network 192.168.40.0 0.0.0.255 area 0

exit

Atur OSPF cost pada interface Cisco.

interface gi0/3

```
description LINK1-TO-HUAWEI
ip ospf cost 10
exit
```

```
interface gi0/0
description LINK2-TO-HUAWEI
ip ospf cost 10
exit
```

```
interface gi0/1
description LINK-TO-MIKROTIK
ip ospf cost 100
exit
```

Verifikasi OSPF Cisco:

```
show ip ospf neighbor
show ip route ospf
```

Hasil kemungkinan kosong karena router tetangga masih belum dikonfigurasi

```
show ip protocols
```

Jika ip vlan tidak muncul, simpan konfigurasi cisco router dengan perintah wr, lalu ketik perintah reload. Tunggu proses restart sampai selesai, lalu cek lagi.

Pastikan network yang akan diiklankan sudah ada:

4. Konfigurasi OSPF pada Huawei Router

Masuk ke huawei router:

```
system-view
```

Konfigurasi OSPF:

```
ospf 1 router-id 2.2.2.2
area 0.0.0.0
network 10.10.10.0 0.0.0.3
network 10.10.11.0 0.0.0.3
network 10.10.20.0 0.0.0.3
network 192.168.50.0 0.0.0.255
quit
quit
```

Atur OSPF cost pada interface huawei.

```
interface Ethernet1/0/0
```

quit

quit

quit

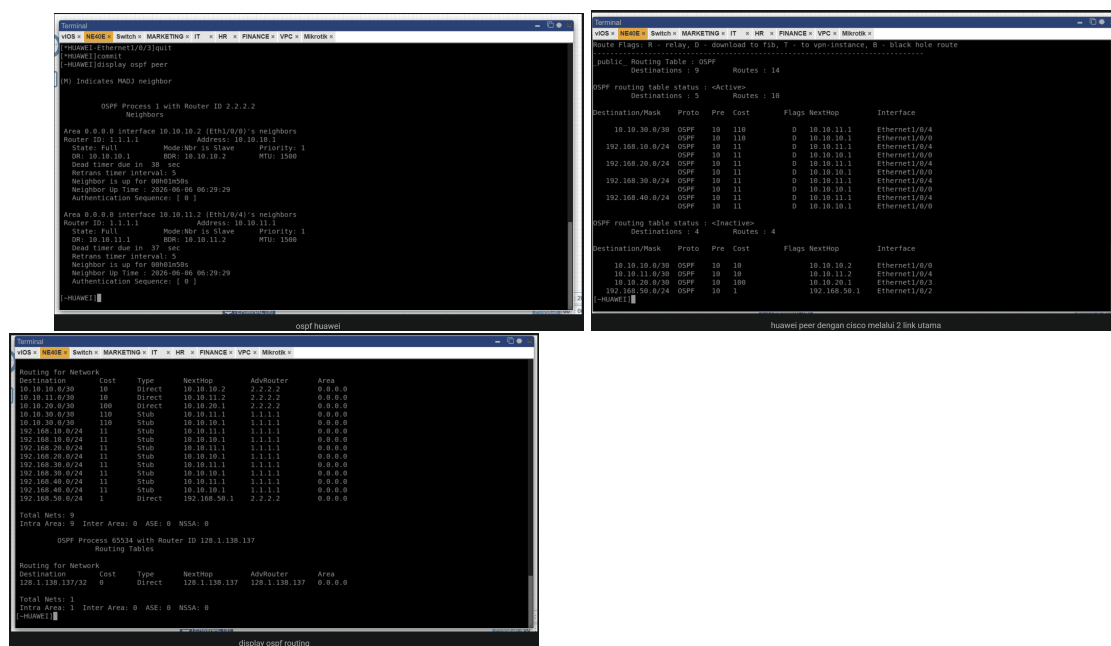
Verifikasi OSPF Huawei:

Pastikan huawei peer dengan cisco melalui 2 link utama.

```
display ip routing-table protocol ospf
```

Pastikan hauwei sudah punya mendapatkan network lengkap dari cisco.

```
display ospf routing
```



5. Konfigurasi OSPF pada MikroTik

Pada topologi ini, Mikrotik digunakan sebagai jalur backup antara cisco dan huawei.

Network Mikrotik:

Link Interface MikroTik IP Address

MikroTik Huawei ether1 10.10.20.2/30

MikroTik Cisco ether2 10.10.30.2/30

Masuk ke terminal Mikrotik, lalu konfigurasi router-id OSPF:

```
/routing ospf instance set default router-id=3.3.3.3
```

Tambahkan network OSPF:

```
/routing ospf network add network=10.10.20.0/30 area=backbone
```

```
/routing ospf network add network=10.10.30.0/30 area=backbone
```

Atur cost interface mikrotik menjadi jalur backup:

```
/routing ospf interface add interface=ether1 cost=100
```

```
/routing ospf interface add interface=ether2 cost=100
```

Verifikasi Mikrotik

```
/routing ospf neighbor print
```

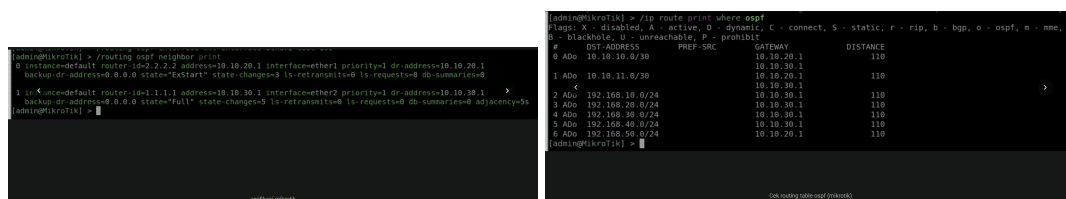
Pastikan router id cisco dan huawei sudah muncul dimikrotik

Cek routing table OSPF:

```
/ip route print where ospf
```

Hasil yang diharapkan:

MikroTik mendapatkan route OSPF menuju VLAN 10, 20, 30, 40 dan LAN Huawei 192.168.50.0/24.



The image contains two terminal screenshots. The left screenshot shows the configuration of OSPF neighbors and the router ID. The right screenshot shows the OSPF routing table.

```
[admin@mikrotik] > /routing ospf neighbor print
# backup-dr address=0.0.0 state="ExStart" state-changes3 ls-retransmits0 ls-requests0 db-summaries0
1 <-> <accedefault router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
# backup-dr address=0.0.0 state="Full" state-changes5 ls-retransmits0 ls-requests0 db-summaries0 adjacency=50
[admin@mikrotik] >

[admin@mikrotik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC      GATEWAY      DISTANCE
0 ADO 10.10.10.0/30  10.10.20.1    110
1 ADO 10.10.11.0/30   10.10.20.1    110
2 ADO 192.168.10.0/24   10.10.30.1    110
3 ADO 192.168.20.0/24   10.10.30.1    110
4 ADO 192.168.30.0/24   10.10.30.1    110
5 ADO 192.168.40.0/24   10.10.30.1    110
6 ADO 192.168.50.0/24   10.10.20.1    110
[admin@mikrotik] >
```

6. Verifikasi OSPF Neighbor Cisco Router

```
show ip ospf neighbor
```

Hasil yang diharapkan:

Neighbor Huawei muncul dua kali.

Neighbor MikroTik muncul satu kali.

Huawei Router

display ospf peer

Hasil yang diharapkan:

Peer ospf muncul dua kali.

Peer mikrotik muncul satu kali.

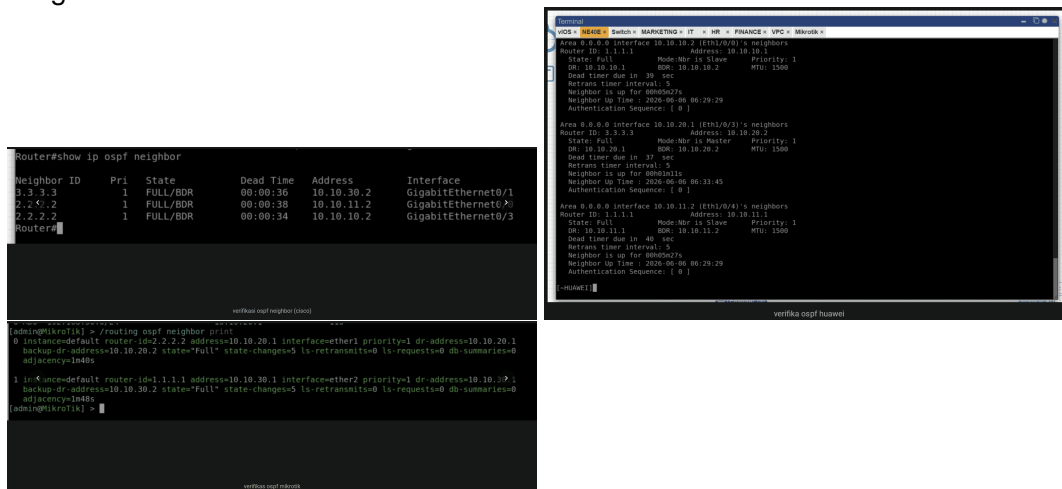
Mikrotik

/routing ospf neighbor print

Hasil yang diharapkan:

Neighbor cisco muncul.

Neighbor Huawei muncul.



7. Verifikasi Routing Table

Cisco Router Cek route menuju lan huawei:

show ip route 192.168.50.0

Hasil yang diharapkan pada kondisi normal:

Jalur menuju network 192.168.50.0/24 melewati cisco-huawei. Jika dua next-hop muncul, artinya ospf menggunakan dua link utama

Cek semua route ospf:

show ip route ospf

Huawei Router

Cek semua route ospf:

display ip routing table protocol ospf

Mikrotik

/ip route print where ospf

Hasil yang diharapkan:

Mikrotik memiliki route ospf ke cisco dan huawei.

```
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:06:18 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:06:18 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:06:18 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#
```

```
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:04:55, GigabitEthernet0/0
O    10.10.20.0/30 [110/110] via 10.10.10.2, 00:04:55, GigabitEthernet0/3
O    192.168.50.0/24 [110/11] via 10.10.11.2, 00:06:40, GigabitEthernet0/0
O    192.168.50.0/24 [110/11] via 10.10.10.2, 00:06:40, GigabitEthernet0/3
Router#
```

```
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:04:55, GigabitEthernet0/0
O    10.10.20.0/30 [110/110] via 10.10.10.2, 00:04:55, GigabitEthernet0/3
O    192.168.50.0/24 [110/11] via 10.10.11.2, 00:06:40, GigabitEthernet0/0
O    192.168.50.0/24 [110/11] via 10.10.10.2, 00:06:40, GigabitEthernet0/3
Router#
```

```
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O    10.10.20.0/30 [110/110] via 10.10.11.2, 00:04:55, GigabitEthernet0/0
O    10.10.20.0/30 [110/110] via 10.10.10.2, 00:04:55, GigabitEthernet0/3
O    192.168.50.0/24 [110/11] via 10.10.11.2, 00:06:40, GigabitEthernet0/0
O    192.168.50.0/24 [110/11] via 10.10.10.2, 00:06:40, GigabitEthernet0/3
Router#
```

8. Verifikasi Koneksi end-to-end

Lakukan pengujian dari pc vlan menuju lan hauwei.

Dari pc vlan 10:

ping 192.168.50.10

Dari pc vlan 20:

ping 192.168.50.10

Dari pc vlan 30:

ping 192.168.50.10

Dari pc vlan 40:

ping 192.168.50.10

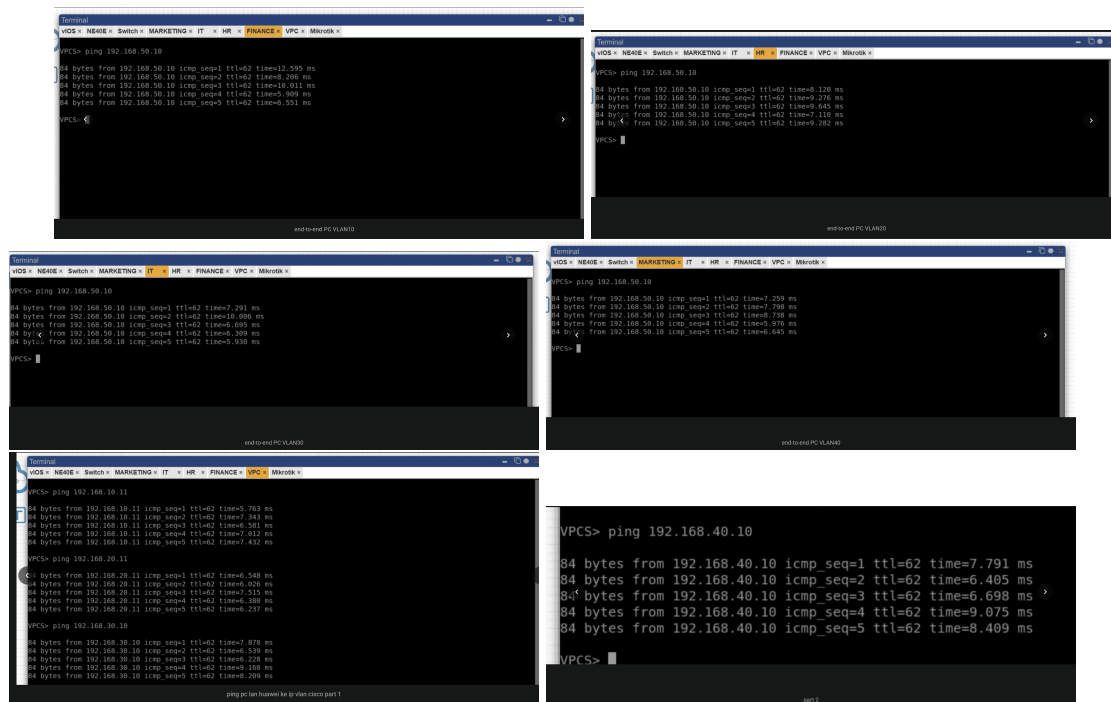
Lakukan pengujian dari pc lan huawei ke ip vlan cisco:

ping 192.168.10.11

ping 192.168.20.11

ping 192.168.30.10

ping 192.168.40.10



1.5 yang kelima

1. Kondisi normal: Dua link cisco-huawei aktif

Cek route dari cisco menuju lan huawei:

```
show ip route 192.168.50.0
```

Lakukan trace dari pc vlan menuju lan huawei:

```
trace 192.168.50.10
```

Hasil yang diharapkan:

Traffic melewati jalur langsung Cisco → Huawei.



2. Kondisi 1 link cisco-huawei mati

Matikan salah satu link cisco-huawei.

```
configure terminal
```

```
interface gi0/3
```

```
shutdown
```

```
exit
```

Cek route dari cisco menuju lan huawei:

```
show ip route 192.168.50.0
```

Lakukan trace dari pc vlan menuju lan huawei:

```
trace 192.168.50.10
```

Hasil yang diharapkan:

Traffic tetap melewati jalur langsung Cisco

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    4.828 ms  6.104 ms  4.002 ms
 2  10.10.11.2     6.361 ms  5.135 ms  3.211 ms
 3  *192.168.50.10 11.337 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

3. Kondisi dua link cisco-huawei mati

Matikan link cisco-huawei yang kedua:

```
configure terminal
```

```
interface gi0/0
```

```
shutdown
```

```
exit
```

Cek route dari cisco menuju lan huawei:

```
show ip route 192.168.50.0
```

Lakukan trace dari pc vlan menuju lan huawei:

```
trace 192.168.50.10
```

Hasil yang diharapkan:

Route berpindah ke jalur backup melalui MikroTik.

Jalur:

PC VLAN → Cisco Router → MikroTik → Huawei Router → LAN Huawei

```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.592 ms  3.801 ms  3.380 ms
 2  10.10.30.2     5.119 ms  3.707 ms  3.572 ms
 3  < 10.10.20.1    8.289 ms  5.543 ms  6.494 ms
 4  *192.168.50.10 12.545 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

4. Mengaktifkan kembali dua link cisco huawei

```
configure terminal
```

```
interface gi0/3
```

```
no shutdown
```

```
exit
```

```
interface gi0/0
```

no shutdown

exit

Cek route dari cisco menuju lan huawei:

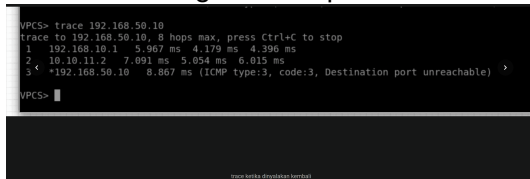
```
show ip route 192.168.50.0
```

Lakukan trace dari pc vlan menuju lan huawei:

```
trace 192.168.50.10
```

Hasil yang diharapkan:

Jalur routing kembali pindah ke link utama cisco-huawei.



```
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl-C to stop
 0  192.168.10.1    5.967 ms  4.172 ms  4.596 ms
 1  10.10.11.2     7.091 ms  5.054 ms  6.015 ms
 2  192.168.50.10  8.867 ms  (ICMP type:3, code:3, Destination port unreachable)
VPCS>
```

2 Analisis Hasil Percobaan

2.1 Analisis Percobaan firewall and nat

pada praktikum 1 Konfigurasi VLAN pada Cisco Switch berjalan sesuai dengan prosedur. VLAN 10 (Finance), 20 (HR), 30 (IT), dan 40 (Marketing) berhasil dibuat dan statusnya aktif, yang dapat diverifikasi melalui perintah show vlan brief. Setiap port access telah dikonfigurasi sesuai VLAN masing-masing, yaitu Gi0/2 untuk VLAN 10, Gi0/1 untuk VLAN 20, Gi1/1 untuk VLAN 30, dan Gi0/3 untuk VLAN 40. Port Gi0/0 berhasil dikonfigurasi sebagai trunk port dengan enkapsulasi dot1Q yang membawa VLAN 10, 20, 30, dan 40 menuju Cisco Router, terkonfirmasi melalui show interfaces trunk. Hal ini sesuai dengan teori bahwa trunk port digunakan untuk membawa lebih dari satu VLAN dalam satu link fisik menggunakan tagging 802.1Q.

pada praktikum 2 Implementasi Router-on-a-Stick (ROAS) pada Cisco Router berhasil dilakukan. Subinterface Gi0/2.10, Gi0/2.20, Gi0/2.30, dan Gi0/2.40 masing-masing dikonfigurasi sebagai gateway untuk tiap VLAN dengan enkapsulasi dot1Q yang sesuai. DHCP Server untuk VLAN 10 dan 20 berfungsi dengan baik, terbukti dari PC VLAN 10 dan 20 berhasil mendapatkan IP secara otomatis melalui perintah ip dhcp, yang dapat diverifikasi lewat show ip dhcp binding. PC VLAN 30 dan 40 menggunakan IP statis sesuai ketentuan. Seluruh PC berhasil melakukan ping ke gateway masing-masing, dan pengujian ping antar-VLAN juga berhasil, membuktikan bahwa inter-VLAN routing melalui router berjalan dengan baik sesuai konsep ROAS.

pada praktikum 3 Konfigurasi IP pada seluruh link antar-router berhasil dilakukan. Link Cisco–Huawei menggunakan dua jalur (10.10.10.0/30 dan 10.10.11.0/30), link Huawei–MikroTik menggunakan 10.10.20.0/30, dan link Cisco–MikroTik menggunakan 10.10.30.0/30. LAN Huawei dengan network 192.168.50.0/24 juga berhasil dikonfigurasi dan PC LAN Huawei dapat melakukan ping ke gateway 192.168.50.1. Perlu dicatat adanya koreksi pada topologi, yaitu Gi0/0 Cisco seharusnya terhubung ke Ethernet1/0/4 Huawei dan Gi0/3 Cisco ke Ethernet1/0/0 Huawei. Setelah penyesuaian tersebut, seluruh router berhasil saling ping ke router tetangga langsung, yang menandakan konektivitas Layer 3 antar-perangkat telah berfungsi dengan benar.

pada praktikum 4 OSPF berhasil dikonfigurasi pada tiga perangkat dari vendor berbeda: Cisco (router-id 1.1.1.1), Huawei (router-id 2.2.2.2), dan MikroTik (router-id 3.3.3.3), semuanya dalam area backbone (area 0). Penetapan OSPF cost sesuai desain: link Cisco–Huawei diberi cost 10 sebagai jalur utama, sedangkan link yang melibatkan MikroTik diberi cost 100 sebagai jalur backup. Cisco Router berhasil membentuk neighbor dengan Huawei melalui dua link dan dengan MikroTik satu link. Huawei berhasil membentuk peer dengan Cisco dan MikroTik. MikroTik pun berhasil mengenali kedua router tetangganya. Seluruh jaringan saling mengetahui route satu sama lain, terbukti dari keberhasilan ping end-to-end antara PC VLAN dan PC LAN Huawei. Hal ini membuktikan bahwa OSPF mampu bekerja secara multivendor selama berada dalam area dan parameter yang sama.

pada praktikum 5 Pengujian failover membuktikan mekanisme redundansi OSPF berjalan sesuai teori. Pada kondisi normal dengan dua link Cisco–Huawei aktif, traffic melewati jalur langsung dengan cost terendah. Saat satu link dimatikan (Gi0/3 di-shutdown), OSPF secara otomatis merutekan traffic melalui link Cisco–Huawei yang tersisa tanpa perlu intervensi manual, menunjukkan konvergensi OSPF yang cepat. Ketika kedua link Cisco–Huawei dimatikan, routing table berpindah ke jalur backup melalui MikroTik (Cisco → MikroTik → Huawei), yang terlihat jelas pada hasil perintah traceroute. Setelah kedua link diaktifkan kembali, OSPF kembali memilih jalur utama Cisco–Huawei secara otomatis. Hal ini sesuai dengan konsep OSPF cost-based path selection dan kemampuan reconvergence-nya.

3 Hasil Tugas Modul

4 Kesimpulan

Praktikum Modul 5 Jarkom berhasil dilaksanakan secara keseluruhan, mulai dari konfigurasi VLAN dan trunk pada Cisco Switch, implementasi ROAS dan DHCP Server pada Cisco Router, pengalaman IP pada seluruh link antar-router multivendor (Cisco, Huawei, MikroTik), hingga penerapan routing dinamis OSPF dengan mekanisme cost untuk menentukan jalur utama dan backup. Seluruh hasil percobaan sesuai dengan teori, di mana OSPF terbukti mampu bekerja lintas vendor dalam satu area backbone, melakukan konvergensi secara otomatis saat terjadi perubahan topologi, serta memastikan konektivitas end-to-end antara PC di jaringan VLAN Cisco dan PC di LAN Huawei tetap terjaga melalui mekanisme failover ke jalur MikroTik ketika kedua link utama Cisco–Huawei terputus.

5 Lampiran

5.1 Dokumentasi saat praktikum